

E. C++ Syntax Reference Guide

Name	Syntax	Example
Pre-processor directives	<pre>#include <libraryName> #define <MACRO_NAME> <expansion></pre>	<pre>#include <iostream> // for CIN & COUT #include <iomanip> // for setw() #include <fstream> // for IFSTREAM #include <string> // for STRING #include <cctype> // for ISUPPER #include <cstring> // for STRLEN #include <cstdlib> // for ATOF #define PI 3.14159 #define LANGUAGE "C++"</pre>
Function	<pre><ReturnType> <functionName>(<params>) { <statements> return <value>; }</pre>	<pre>int main() { cout << "Hello world\n"; return 0; }</pre>
Function parameters	<pre><DataType> <passByValueVariable>, <DataType> & <passByReferenceVariable>, const <DataType> <CONSTANT_VARIABLE>, <BaseType> <arrayVariable>[]</pre>	<pre>void function(int value, int &reference, const int CONSTANT, int array[]) { }</pre>
COUT	<pre>cout << <expression> << ... ;</pre>	<pre>cout << "Text in quotes" << 6 * 7 << getNumber() << endl;</pre>
Formatting output for money	<pre>cout.setf(ios::fixed); cout.setf(ios::showpoint); cout.precision(<integerExpression>);</pre>	<pre>cout.setf(ios::fixed); cout.setf(ios::showpoint); cout.precision(2);</pre>
Declaring variables	<pre><DataType> <variableName>; <DataType> <variableName> = <init>; const <DataType> <VARIABLE_NAME>;</pre>	<pre>int integerValue; float realNumber = 3.14159; const char LETTER_GRADE = 'A';</pre>
CIN	<pre>cin >> <variableName>;</pre>	<pre>cin >> variableName;</pre>
IF statement	<pre>if (<Boolean-expression>) { <statements> } else { <statements> }</pre>	<pre>if (grade >= 70.0) cout << "Great job!\n"; else { cout << "Try again.\n"; pass = false; }</pre>
Asserts	<pre>assert(<Boolean-expression>);</pre>	<pre>#include <cassert> // at top of file { assert(gpa >= 0.0); }</pre>

Name	Syntax	Example
FOR loop	<pre>for (<initialization statement>; <Boolean-expression>; <increment statement>) { <statements> }</pre>	<pre>for (int iList = 0; iList < sizeList; iList++) cout << list[iList];</pre>
WHILE loop	<pre>while (<Boolean-expression>) { <statements> }</pre>	<pre>while (input <= 0) cin >> input;</pre>
DO-WHILE Loop	<pre>do { <statements> } while (<Boolean-expression>);</pre>	<pre>do cin >> input; while (input <= 0);</pre>
Read from File	<pre>ifstream <streamVar>(<fileName>); if (<streamVar>.fail()) { <statements> } <streamVar> >> <variableName>; <streamVar>.close();</pre>	<pre>#include <fstream> // at top of file { ifstream fin("data.txt"); if (fin.fail()) return false; fin >> value; fin.close(); }</pre>
Write to File	<pre>ofstream <streamVar>(<fileName>); if (<streamVar>.fail()) { <statements>; } <streamVar> << <expression>; <streamVar>.close();</pre>	<pre>#include <fstream> // at top of file { ofstream fout("data.txt"); if (fout.fail()) return false; fout << value << endl; fout.close(); }</pre>
Fill an array	<pre><BaseType> <arrayName>[<size>]; <BaseType> <arrayName>[] = { <CONST_1>, <CONST_2>, ... }; for (int i = 0; i < <size>; i++) <arrayName>[i] = <expression>;</pre>	<pre>int grades[10]; for (int i = 0; i < 10; i++) { cout << "Grade " << i + 1 << ": "; cin >> grades[i]; }</pre>
C-Strings	<pre>char <stringName>[<size>]; cin >> <stringName>; for (char * <ptrName> = <stringName>; *<ptrName>; <ptrName>++) cout << *<ptrName>;</pre>	<pre>char firstName[256]; cin >> firstName; for (char * p = firstName; *p; p++) cout << *p;</pre>
String Class	<pre>string <stringName>; cin >> <stringName>; cout << <stringName>; getline(<streamName>, <stringName>); if (<stringName1> == <stringName2>) <statement>; <stringName1> += <stringName2>; <stringName1> = <stringName2>;</pre>	<pre>string string1; // declare string string2 = "124"; // initialize cin >> string1; // input getline(cin, string2); // getline if (string1 == string2) // compare string1 += string2; // append string2 = string1; // copy</pre>

Name	Syntax	Example
Switch	<pre>switch (<integer-expression>) { case <integer-constant>: <statements> break; // optional ... default: // optional <statements> }</pre>	<pre>switch (value) { case 3: cout << "Three"; break; case 2: cout << "Two"; break; case 1: cout << "One"; break; default: cout << "None!"; }</pre>
Conditional Expression	<pre><Boolean-expression> ? <expression> : <expression></pre>	<pre>cout << "Hello, " << (isMale ? "Mr. " : "Mrs. ") << lastName;</pre>
Multi-dimensional array	<pre><BaseType> <arrayName>[<SIZE>][<SIZE>]; <BaseType> <arrayName>[][<SIZE>] = { { <CONST_0_0>, <CONST_0_1>, ... }, { <CONST_1_0>, <CONST_1_1>, ... }, ... }; <arrayName>[<index>][<index>] = <expression>;</pre>	<pre>int board[3][3]; for (int row = 0; row < 3; row++) for (int col = 0; col < 3; col++) board[row][col] = 10;</pre>
Allocate memory	<pre><ptr> = new(nothrow) <DataType>; <ptr> = new(nothrow) <DataType>(<init>); <ptr> = new(nothrow) <BaseType>[<SIZE>;</pre>	<pre>float * p1 = new(nothrow) float; int * p2 = new(nothrow) int(42); char * text = new(nothrow) char[256];</pre>
Free memory	<pre>delete <pointer>; // one value delete [] <arrayPointer>; // an array</pre>	<pre>delete pNumber; delete [] text;</pre>
Command line parameters	<pre>int main(int <countVariable>, char **<arrayVariable>) { }</pre>	<pre>int main(int argc, char ** argv) { }</pre>

Library	Function Prototype
#include <cctype>	<pre>bool isalpha(char); // is the character an alpha ('a' - 'z' or 'A' - 'Z')? bool isdigit(char); // is the character a number ('0' - '9')? bool isspace(char); // is the character a space (' ' or '\t' or '\n')? bool ispunct(char); // is the character a symbol such as %#\$!-_*@.,? bool isupper(char); // is the character uppercase ('A' - 'Z')? bool islower(char); // is the character lowercase ('a' - 'z')? int toupper(char); // convert lowercase character to uppercase. Rest unchanged int tolower(char); // convert uppercase character to lowercase. Rest unchanged</pre>
#include <cstring>	<pre>int strlen(const char *); // find the length of a c-string int strcmp(const char *, const char *); // 0 if the two strings are the same char * strcpy(char * <dest>, const char * <src>); // copies src onto dest</pre>
#include <cstdlib>	<pre>double atof(const char *); // parses input for a floating point number and returns it int atoi(const char *); // parses input for an integer number and returns it</pre>